

Induction

Let us look again, from a certain angle, at the problem of “induction.” It becomes one step beyond archaic in a modern environment, making the Black Swan problem even more severe. Simply, in a complex domain, the discussion of induction versus deduction becomes too marginal to the real problems (except for a limited subset of variables, even then); the entire Aristotelian distinction misses an important dimension (similar to the one discussed earlier concerning the atypicality of events in Extremistan). Even other notions such as “cause” take on a different meaning, particularly in the presence of circular causality and interdependence.* The probabilistic equivalent is the move from a conventional random walk model (with a random variable moving in a fixed terrain and not interacting with other variables around it), to percolation models (where the terrain itself is stochastic, with different variables acting on one another).

Driving the School Bus Blindfolded

Alas, at the time of writing, the economics establishment is still ignorant of the presence of complexity, which degrades predictability. I will not get too involved in my outrage—instead of doing a second *deserto*, Mark Spitznagel and I are designing another risk management program to robustify portfolios against model error, error mostly stemming from the government’s error in the projection of deficits, leading to excessive borrowing and possible hyperinflation.

I was once at the World Economic Forum in Davos; at one of my sessions, I illustrated interdependence in a complex system and the degradation of forecasting, with the following scheme: unemployment in New York triggered by Wall Street losses, percolating and generating unemployment in, say, China, then percolating back into unemployment in New York, is not analyzable analytically, because the feedback loops produced monstrous estimation errors. I used the notion of “convexity,” a disproportionate nonlinear response stemming from a variation in input (as the tools for measuring error rates go out of the window in the presence of convexity). Stanley Fisher, the head of the central bank of Israel, former IMF hotshot, co-author of a classic macroeconomics textbook, came to talk to me after the session to critique my point about such feedback loops causing unpredictability. He explained that we had input-output matrices that were good at calculating such feedbacks, and he cited work honored by the “Nobel” in economics. The economist in question was one Vassili Leontieff, I presume. I looked at him with the